AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	Vantage Data Centers TX22, TX -- station KSAT, America/Chicago
Committed pair	OSM 509611432 -> 509611429 CyrusOne San Antonio I / OSM way 509611429
Facade gap	271.8 m between the two halls
Weather record	43,809 real hourly records from KSAT
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.2929 C at 180 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-01-07 (safe_sector)
Plant limit	18.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	1 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 7 of 24 hours with 1 mode change. The reactive incumbent took 13 hours -- 6 more -- but declared 0 unsafe hours against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 13 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	7 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	13 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s) the incumbent broke its own switch budget 1 time(s) to stay safe 13 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	10.106	18.0	10.000	1.168
01	mechanical	yes	switch budget	9.556	18.0	8.890	1.017
02	mechanical	yes	switch budget	9.235	18.0	8.330	0.838
03	mechanical	yes	switch budget	9.482	18.0	8.890	0.669
04	mechanical	yes	switch budget	8.987	18.0	8.330	0.869
05	mechanical	yes	switch budget	8.130	18.0	7.220	0.769

06	mechanical	yes	switch budget	8.662	18.0	7.780	0.646
07	mechanical	yes	switch budget	8.661	18.0	7.780	0.621
08	mechanical	yes	switch budget	10.409	18.0	9.440	1.240
09	mechanical	yes	switch budget	13.662	18.0	11.670	2.203
10	mechanical	yes	switch budget	15.022	18.0	13.330	2.146
11	mechanical	yes	switch budget	15.880	18.0	15.000	1.406
12	mechanical	yes	switch budget	16.983	18.0	16.110	1.049
13	mechanical	no	dry-bulb	18.127	18.0	17.220	1.069
14	mechanical	no	dry-bulb	18.977	18.0	17.780	1.145
15	mechanical	no	dry-bulb	19.250	18.0	18.330	1.031
16	mechanical	no	dry-bulb	18.639	18.0	17.220	0.967
17	FREE	yes	-	16.978	18.0	14.440	1.111
18	FREE	yes	-	15.864	18.0	13.330	1.018
19	FREE	yes	-	13.384	18.0	10.000	1.385
20	FREE	yes	-	10.595	18.0	7.780	1.350
21	FREE	yes	-	9.473	18.0	6.110	1.527
22	FREE	yes	-	8.809	18.0	7.864	1.344
23	FREE	yes	-	7.526	18.0	7.780	0.988

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.106 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.557 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.235 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.482 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.987 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.130 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.663 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.661 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.409 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.663 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.022 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

11:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.880 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

12:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.983 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.127 C against a 18.0 C limit, so it fails by 0.127 C. It would take a limit 0.127 C higher, or a bound 0.127 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.978 C against a 18.0 C limit, so it fails by 0.978 C. It would take a limit 0.978 C higher, or a bound 0.978 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.250 C against a 18.0 C limit, so it fails by 1.250 C. It would take a limit 1.250 C higher, or a bound 1.250 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.639 C against a 18.0 C limit, so it fails by 0.639 C. It would take a limit 0.639 C higher, or a bound 0.639 C tighter, to change this hour.

17:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.978 C, which is 1.022 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.111 C, and the margin is measured, not chosen: 1.078 C of group-conditional forecast error for this hour of day, plus 0.0329 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.864 C, which is 2.136 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.018 C, and the margin is measured, not chosen: 0.984 C of group-conditional forecast error for this hour of day, plus 0.0339 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 13.384 C, which is 4.616 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.385 C, and the margin is measured, not chosen: 1.331 C of group-conditional forecast error for this hour of day, plus 0.0541 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.595 C, which is 7.405 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.350 C, and the margin is measured, not chosen: 1.311 C of group-conditional forecast error for this hour of day, plus 0.0396 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.473 C, which is 8.527 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.527 C, and the margin is measured, not chosen: 1.493 C of group-conditional forecast error for this hour of day, plus 0.0332 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.809 C, which is 9.191 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.344 C, and the margin is measured, not chosen: 1.228 C of group-conditional forecast error for this hour of day, plus 0.1156 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.526 C, which is 10.474 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.988 C, and the margin is measured, not chosen: 0.967 C of group-conditional forecast error for this hour of day, plus 0.0213 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

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